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Police Fairness, Anger and Support for Violent and Non-Violent Disruptive Actions in Chile

Micaela Varela[1] | Rezarta Bilali[1] | Joaquin Bahamondes[2] | Ana Figueiredo[3] | Monica M. Gerber[4,5] | Erin Godfrey[1]

[1]Department of Applied Psychology, New York University, New York, USA | [2]School of Psychology, Universidad Católica del Norte, Antofagasta, Chile | [3]Institute of Social Sciences, Universidad de O'Higgins, Rancagua, Chile | [4]Faculty of Psychology, Universidad Diego Portales, Santiago, Chile | [5]School of Government, Pontificia Universidad Católica de Chile, Santiago, Chile

Correspondence: Micaela Varela (micaela.varela@nyu.edu)

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ABSTRACT

Cross-sectional research has shown that perceptions of injustice and their resultant emotions are associated with increased support for collective actions. In a three-wave longitudinal study of Chilean citizens ($N_{T1}=2611$; $N_{T2}=1690$; $N_{T3}=1163$), using Random Intercept Cross-Lagged Panel models, we examined the associations over time between perceived procedural justice of the police during protests, anger towards the police, and justification for non-violent and violent disruptive actions during a period of de-escalation of protests. Our analyses revealed that people who reported perceiving higher procedural justice at Time 1 (January 2021, 2 months after the referendum to change the constitution) expressed lower anger towards the police and lower justification of protesters' violent actions in Time 2 (August 2021, after the election of Constitutional representatives). These associations, however, were not significant between Time 2 and Time 3 (January 2022, following the election of Boric as president of Chile). Surprisingly, higher levels of anger towards the police in Time 2 predicted lower support for violent disruptive actions in Time 3. We interpret these results by considering the dynamic de-escalation of the protests and the ongoing sociopolitical changes in Chile, highlighting the relevance of the context in the study of collective action.

1 | Introduction

Collective action and protests are often met with repression and police violence (e.g., Turkey 2013, Chile 2019, Iraq 2019–2020, Hong Kong 2019–2020, Black Lives Matter protests 2020, Colombia 2021, among other examples across the world). A growing body of literature on collective action suggests that the police's unfair treatment of protesters, especially violent repression, is associated with higher expressions of dissent (Curtice and Behlendorf 2021), motivation for engaging in collective action (Adam-Troian et al. 2020; Gerber et al. 2021; Stott et al. 2020), and support for and motivation to engage in anti-government and anti-police violence by protesters (e.g., Bartusevičius et al. 2023; Gerber et al. 2023; Snipes et al. 2019; Tyler et al. 2018). Perceptions of police behaviour as unfair, repressive, or violent can evoke anger towards authorities (Ayanian and Tausch 2016; Maguire et al. 2023) and contribute to the politicisation and radicalization of protesters' and observers' social identities (e.g., Adam-Troian et al. 2020; Drury and Reicher 2009; Reicher and Stott 2020). The use of excessive force to suppress protest can intensify anger and perceived injustice, which in turn can feed the escalation of violence, leading to more extreme actions taken by protestors like throwing stones towards the police or damaging public infrastructure (Drury and Reicher 2000; Reicher et al. 2004; Smith et al. 2020; but also see Davenport 2007, and Lichbach 1987 on how the use of excessive force and repression might be effective in de-escalating or stopping collective action altogether).

Much of the literature on police violence and collective action has focused on theorising and understanding the escalating dynamics of confrontation between the police and protesters (Drury and Reicher 2000; Reicher et al. 2004; Tyler et al. 2018). However, collective action and social movements can also contribute to large-scale social and institutional transformation. Research in this area often overlooks the evolving socio-political contexts in which protests unfold, in part due to the predominance of cross-sectional designs (Louis and Montiel 2018). To advance this understanding, it is essential to consider not only the strategies of protesters and their interactions with police—actors embedded in asymmetrical power relations—but also the broader processes of political contention specific to a given time and place (Demetriou et al. 2014). When protests lead to structural and institutional changes, these shifts may alter the dynamics between state actors and the public, potentially resulting in de-escalation or a decline in protest activity. Yet, existing research has not sufficiently examined how the relationships between procedural justice, anger, and support for collective action may shift during periods of major socio-political change. This raises a critical question: do these relationships change in times of structural transformation?

We examined these dynamics in the aftermath of the 2019 Chilean social uprising, which began in October with protests against a subway fare increase. These protests rapidly escalated into periodic protests about different social grievances and against the right-wing government of Sebastian Piñera. They were met with heavy police repression and societal unrest (D'Ottone et al. 2024; Malone and Dammert 2021; Zúñiga et al. 2021). A month into the protests, in a desperate move to de-escalate the situation in the country, the national congress signed an agreement to hold a referendum to create a new constitution—a key demand of the protesters. Protests continued but were somewhat diminished due to COVID-19 pandemic restrictions on public gatherings (Donoso et al. 2022; López-Contreras et al. 2022). The referendum was held in October 2020, resulting in an overwhelming vote in favour of changing the constitution (78.28% voted in favour). Since the beginning of the protests, Chile engaged in fast-paced structural changes—including a very long lockdown during the pandemic, a constitutional process, and presidential elections in which Gabriel Boric, a 36-year-old leftist candidate, was elected president—while simultaneously grappling with revelations of previous police corruption and human rights violations (Ruiz 2020; Malone and Dammert 2021).

This study addresses key gaps in the literature by leveraging a three-wave longitudinal design to examine changes in perceived police procedural justice, anger, and support for disruptive collective action in post-uprising Chile. It contributes in three main ways: first, by moving beyond cross-sectional designs to explore within- and between-person change over time; second, by distinguishing between violent and non-violent disruptive actions, offering a more nuanced understanding of protest tactics; and third, by analysing a context of de-escalation to assess whether established theories of protest dynamics hold as the socio-political status quo evolves.

2 | Procedural Justice and Support for Violent and Non-Violent Disruptive Collective Action

The perception of injustice is a key catalyst for collective action, uniting people who share grievances and aspire to social change (Lizzio-Wilson et al. 2022; Louis et al. 2022; van Zomeren et al. 2012). The role of perceived grievances has been widely recognised by the social identity model of collective action (SIMCA, van Zomeren et al. 2008). However, one form of grievance that has received little attention is procedural injustice (e.g., Van Stekelenburg and Klandermans 2013; Zhu and Chou 2021).

Depending on how injustice is perceived, the individual motivations and the contextual factors at play, people can support different types of collective action, such as non-violent and violent strategies (Becker and Tausch 2015; Selvanathan 2019). The distinction between non-violent and violent action varies across individuals, groups, cultures, and historical moments (Drury and Reicher 2018; Uysal et al. 2024). In this article, we focus on disruptive collective action and distinguish between violent and non-violent disruptive collective action. In contrast to non-disruptive actions, such as peaceful demonstrations, disruptive actions obstruct the day-to-day operation of a city or community (e.g., by closing streets and affecting public transportation), but they are not necessarily violent. During social uprisings and prolonged protests, tactics such as weekly marches or street blockades—though not necessarily approved by authorities—can become accepted and supported by the broader public. Violent actions, by contrast, are more confrontational (Uysal et al. 2024), involving extreme or radical behaviours. Such actions include aggression or physical violence and typically emerge when conventional avenues for addressing injustice are perceived as unavailable or ineffective (Wright et al. 1990; Pauls et al. 2021).

During protests, the interactions between police and protesters play a significant role in shaping perceptions and behaviours (Drury and Reicher 2018; Stott and Drury 2000). Procedural justice refers to the perception that authorities—particularly the police—act fairly during decision-making processes (Lind and Tyler 1988; T. Tyler 2003; Walters and Bolger 2018). When authorities are seen as fair, people are more likely to legitimise them and internalise a sense of duty to obey the law (T. R. Tyler 1990, 2003; Tyler and Huo 2002).

During ongoing protests, when the police are perceived as acting unfairly, protesters are less likely to comply with orders and more likely to express their discontent, either by engaging in or supporting more violent actions (Nassauer 2016; Snipes et al. 2019; Tyler et al. 2018). For example, in interviews with participants from the Occupy Wall Street movement in New York City, Maguire et al. (2018) found that higher perceptions of procedural injustice—such as false arrests or excessive force—were associated with increased support for violence against the police. A comparative study of protests in Portland, Hong Kong, and Santiago similarly found that police repression provoked backlash, increasing destructive behaviours among protesters in all three cities (Maguire 2021). Beyond protesters' perspectives, public perceptions show similar patterns: unfair treatment by authorities—such as political repression—can galvanise broader support for violent collective action (Gerber et al. 2021; Tyler et al. 2018; Zhu et al. 2022) and even political violence (Bartusevičius et al. 2023; Linke et al. 2015).

Much of the research in this area has focused on the link between procedural justice and support for or participation in *violent* actions. One study in Uganda (Curtice and Behlendorf 2021) found that procedural injustice by the police (i.e., excessive use of force) led to backlash and expression of dissent among the public, independent of whether people supported the government or the protesters. Using a representative sample in Chile, Gerber et al. (2021) found that perceiving the authorities as unfair was related to higher support for both violent and non-violent disruptive actions. Zhu et al. (2022) found a complex relationship between the severity of police repression and online public support for violent collective action, such that police repression that is proportional to violence by protestors reduces people's support for protesters' actions. In contrast, disproportional police repression increases the public's justification of protest violence.

The cross-sectional research linking procedural justice with support for violent and non-violent disruptive actions typically assumes a directional influence from police procedural injustice to support of collective action (e.g., Maguire et al. 2018; Tyler et al. 2018; Gerber et al. 2023). In the present study, we built on this research by longitudinally examining the perceptions of the larger public (i.e., including people who support and oppose protests) regarding the police's procedural justice and support for different forms of violent and non-violent disruptive action during a period of protest de-escalation and socio-political changes. We situated our study in the Chilean context in the aftermath of the 2019 social uprising. According to a survey conducted by CASEN (2024), public support for disruptive violence was widespread during the 2019 protests in Chile, with 55% of a representative sample reporting support for them. During August 2021, this support was still sustained, but at a lower level (39%), presenting a de-escalation scenario that responded to contextual changes.

Whether perceptions of police fairness continue to influence the justification of different forms of collective action (and vice versa) over time is an important question because swaying the public's opinion in one direction will likely influence escalation and de-escalation dynamics as well. After the protesters' success in influencing a shift from right-wing to left-wing governance—and therefore, a change in who controls institutions, including the police—the police and their actions might now be viewed in a different light, now that power has shifted. However, the reverse link is also plausible, especially regarding violent actions. Suppose the general public perceives that many protest demands have been met and that elections are signalling a shift in the country (i.e., with the announcement of a new constitution or a progressive president). In that case, people might not support continued protests, leading to viewing police actions (including excessive use of force) towards protesters as fair and necessary (i.e., more perceived procedural justice of the police). Under such circumstances, continued collective action, especially violent action, might be perceived as spoilers of the process, which in turn might increase perceived fairness of police actions, including the use of potentially excessive force.

3 | Anger and the Association Between Procedural Justice and Collective Action

Anger at perceived injustice is a key factor in various collective action models. As proposed by the Social Identity Model of Collective Action (or SIMCA; see van Zomeren et al. 2008), 'when group-based inequality or deprivation is perceived as unjust, group-based emotions like anger should motivate collective action because they invoke specific action tendencies to confront those responsible in order to redress their unfair deprivation' (p. 506). Different meta-analyses have supported this association between perceived or felt injustice and group-based anger (Agostini and van Zomeren 2021; van Zomeren et al. 2008).

While the link between anger and collective action is well established, there are mixed findings regarding the specific types of collective action with which anger is most strongly associated. Some studies have shown that anger predicts support for peaceful collective actions (Ayanian and Tausch 2016; Ayanian et al. 2021; Leach et al. 2006; Saab et al. 2015; Sabucedo et al. 2011; Shuman et al. 2016; van Zomeren et al. 2004). Others have provided evidence for the link between anger and support for violent actions (Fink et al. 2022; Stürmer and Simon 2009; Zhu et al. 2022), and the link between anger and both non-violent and violent actions (Tausch et al. 2011; Zlobina and González Vázquez 2018; Zúñiga et al. 2023).

We use longitudinal data to examine changes within the same individuals regarding how they perceive the police, the felt anger towards the police, and the extent to which they justify non-violent and violent disruptive actions in a context of de-escalation of protests, institutional announcements towards progressive social change, and political transition.

4 | The Current Study

We use three waves from a longitudinal survey on police legitimacy collected by Observatorio de Violencia y Legitimidad Social (OLES). The first wave was collected in January 2021, approximately 3 months after the referendum that approved the writing of a new constitution. The second wave was collected in August 2021, 3 months after the election of the representatives who would draft the new constitution. The third and last wave was collected 3 months after Gabriel Boric was elected president in November. Using these three time points, we explore how the perception of police fairness (procedural justice) is related to anger towards the police and the justification of non-violent and violent disruptive collective action over time.

In October 2019, a subway fare increase in Santiago sparked a long period of protests, referred to as *estallido social* or *revuelta social* (social outbreak or social revolt). The demonstrations swiftly expanded to all 16 regions, culminating in over 3300 protests in just two and a half months, as protesters pushed for changes in labour, education, health, pensions and a New Constitution (Somma et al. 2020; Zúñiga et al. 2021). The right-wing President of Chile at the time, Sebastián Piñera, declared a State of Emergency, which led to curfews and the deployment of military personnel across Chilean cities. Police-protester confrontations were common. Although most protests were peaceful, violent events were reported in 40% of the protests (Malone and Dammert 2021). The weekly clashes between police and protesters resulted in mass injuries, cases of torture, and deaths (INDH 2019).

In a desperate move to de-escalate the situation, on November 15, 2019, the government and several political parties announced an agreement for a referendum, in which people voted on whether they agreed or rejected the idea of drafting a new Constitution to replace the current one, written during Augusto Pinochet's dictatorship. There was a decrease of almost 60% in the daily number of protests in the following weeks after the agreement (Corvalán and Pardow 2020). Notably, the rate of state coercion, measured by the number of arrests, remained constant during said period. The lockdowns during the COVID-19 pandemic also slowed the protests but did not end them (Donoso et al. 2022). The contempt towards Sebastian Piñera's government and authorities persisted, motivating protest support and participation despite health risks. A year after the start of the demonstrations, on October 25, 2020, a referendum was held to write a new constitution, and the majority (78.28%) supported drafting a new constitution. This outcome and a government approval barely reaching 10% gave the left-wing opposition an optimistic overconfidence (El Mercurio 2021, as cited in Mayorga 2024).

A constitutional assembly was elected in March 2021 by popular vote, with a majority of independent and progressive representatives, as well as reserved seats for Indigenous leaders, and ensuring gender parity (Mayorga 2024). However, the assembly soon faced controversy, most notably when one member, Rodrigo Rojas Vade, who had gained public support through his activism, was exposed for lying about his cancer diagnosis, and subsequently resigned from his position as vice president of the convention (Radovic and Chernin 2021). This, along with other issues, undermined the assembly's legitimacy and deepened public distrust in political institutions. By the end of 2021, only 24% of the public reported trusting the work of the constitutional assembly representatives, while trust in the police had increased to 26%, up from a critical low of 17% in late 2019 (Centro de Estudios Públicos 2021). Boric's victory in the December 2021 presidential election led to the announcement of an agenda aligned with the demands for social change and the intention to shift the country's institutional direction.

The data used in this research were collected following the first constitutional referendum in January 2021 (wave 1), then in August 2021 (wave 2), and again in January 2022 (wave 3). The first constitutional convention was held between the first and the second wave of data collection. Between the second and the third wave of data collection, in December 2021, Gabriel Boric was elected as the new President of Chile, bringing a new progressive political group to the government. However, he began his presidency in March 2022, 3 months after the third wave was collected. That is, between wave 1 and wave 2 of the data collection, a right-wing party was governing, and from wave 2 to wave 3 of the data collection, a left-wing party was elected (see Figure 1). This contextual setup allows us to examine within-person changes in a dynamic context of political transition.

Building on prior research, we propose exploratory hypotheses about disruptive actions in general, without distinguishing between violent and non-violent actions. We acknowledge that the mechanisms for non-violent disruptive actions may differ from those of violent disruptive actions. However, we don't make specific predictions about each type of action due to the mixed evidence regarding the relation of anger and non-violent collective action versus violent collective action (Tausch et al. 2011).

H1. *(Procedural Justice → Anger) Higher perceived procedural justice of the police at an earlier time point* (T1 *or* T2) *will be associated with lower anger towards the police at a subsequent time point* (T2 *or* T3).

H2. *(Procedural Justice → Collective Action) Higher perceived procedural justice of the police at an earlier time point* (T1 *or* T2) *will be associated with lower support for disruptive actions at a subsequent time point* (T2 *or* T3).

H3. *(Anger → Collective Action) Lower anger towards the police at an earlier time point* (T1 *or* T2) *will be associated with lower support for disruptive actions at a subsequent time point* (T2 *or* T3).

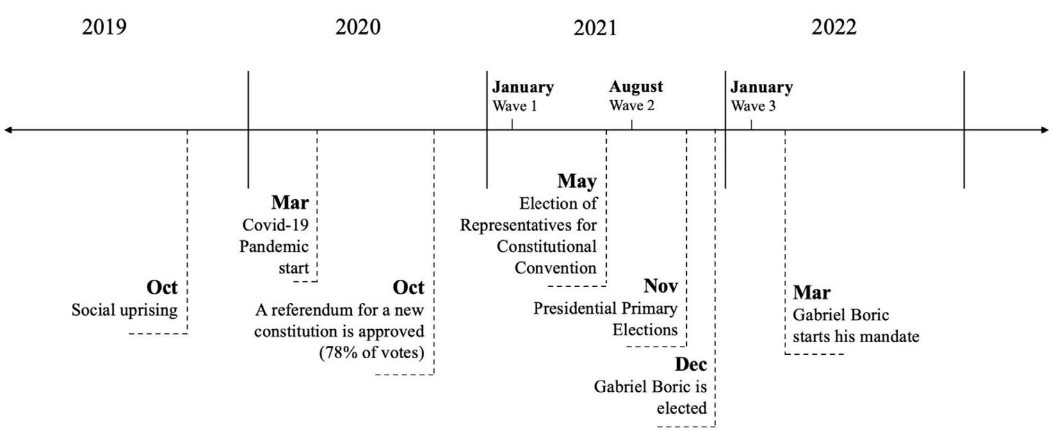

FIGURE 1 | Chilean context and timeline of the data collection.

5 | Method

5.1 | Participants and Procedure

The data was collected by the Observatorio de Violencia y Legitimidad Social (OLES). The study consists of an online longitudinal survey with three waves, collected in January 2021 ($N_{T1} = 2611$), August 2021 ($N_{T2} = 1690$, 65% of participants in Time 1), and January 2022 ($N_{T3} = 1163$, 44% of participants in Time 1).

The sampling was non-probabilistic, using quotas to ensure a balanced representation based on sex (men and women), age group (18–24 years, 25–34 years, 35–44 years, 45–54 years and 55 years and over), and socioeconomic level (high, medium-high, medium-low and low). Netquest collected the data and included Chilean residents from all 16 of Chile's regions.

Of the 2611 participants in Wave 1 ($M_{age} = 39.4$, $SD = 14.4$), 51% identified as women ($N = 1337$). Regarding political orientation, 20% identified with the left and centre-left, 14% with the centre, 10% with the right and centre-right, and 55% indicated that they did not identify with any of the above. Regarding education level, 19% of the respondents reported having completed high school, 18% had completed technical education, and 26% had completed professional education. Slightly less than half of the sample in Wave 1 ($N = 1136$, 44%) reported participating in protests since the beginning of the demonstrations on October 19, 2019.

We examined the patterns of missing data across the three waves. Little's MCAR test for missing data was significant ($\Delta\chi^2$ $(574) = 818$, $p < 0.001$), suggesting that the data is not missing completely at random (MCAR). We conducted a logistic regression analysis to examine demographic factors (specifically, sex, age, political orientation, and SES) associated with the likelihood of missing data in Wave 2 and Wave 3. Women (Wave 2: $\beta = 0.213$, $SE = 0.082$, $p = 0.009$; Wave 3: $\beta = 0.24$, $SE = 0.08$, $p = 0.004$), higher age (Wave 2 and Wave 3: $\beta = 0.02$, $SE = 0.003$, $p < 0.001$), and higher socio-economic levels (Wave 2: $\beta = 0.23$, $SE = 0.04$, $p < 0.001$; Wave 3: $\beta = 0.20$, $SE = 0.04$, $p < 0.001$) had higher odds of participating in subsequent waves. In addition, respondents who reported no political orientation in Wave 1 had lower odds of participation in Wave 3 than participants who identified themselves in the centre of the political spectrum ($\beta = -0.258$, $SE = 0.121$, $p = 0.034$). Considering that the data are not missing completely at random (MCAR), we treated the missing data using Full Information Maximum Likelihood (FIML), incorporating all available information. This strategy yields less biased parameter estimates and more reliable results under the MAR assumption than other conventional methods for handling missing data, such as imputation (Muthén et al. 2017).

5.2 | Measures

The measures used in this study were part of a larger questionnaire. All measures were assessed in all three waves (for more information about the complete questionnaire, see https://oles2.netlify.app/study).

5.2.1 | Support for Violent and Non-Violent Disruptive Collective Action

Seven items measured justification for different disruptive actions during the protests. Participants were asked, 'To what extent do you think the following actions that some protesters may carry out in the context of protests are justified or not justified?' (1 = never justified; 5 = always justified). The actions assessed included: 'holding a demonstration even when it is not authorized', 'disobeying police orders to disperse when protesting', 'blocking streets even when police ordered not to', 'occupying public buildings', 'throwing stones at the police', 'damaging public infrastructure (e.g., traffic lights, signage, streets, buildings)', and 'damaging infrastructure belonging to large companies'.

We conducted an Exploratory Factor Analysis (EFA) with principal axis factoring extraction and oblimin rotation on the seven collective action items in Wave 1 to examine whether these items constitute one or more factors. The analyses revealed two factors, explaining 37.8% and 28.3% of the variance, respectively. The four items loading on the first factor (factor loadings ranging from 0.53 to 0.87) included disruptive but non-violent forms of collective action: 'holding a demonstration even when it is not authorized', 'disobeying police orders to disperse when protesting,' 'blocking streets even when police ordered not to,' and 'occupying public buildings.' The three items loading onto the second factor (factor loadings ranged from 0.67 to 0.85) included disruptive and violent actions: 'throwing stones at the police', 'damaging public infrastructure (e.g., traffic lights, signage, streets, buildings)', and 'damaging infrastructure belonging to large companies'.

Following these analyses, we conducted confirmatory factor analyses (CFA) to examine whether the two-factor model of collective action represented the data across all three waves. The two-factor CFAs revealed good fits across all waves (Wave 1: $CFI = 0.974$, $TLI = 0.958$, $SRMR = 0.034$, $RMSEA = 0.0981$, $AIC = 45{,}424$, $BIC = 45{,}512$, $\chi^2(13) = 291.545$, $p < 0.001$; Wave 2: $CFI = 0.979$, $TLI = 0.966$, $SRMR = 0.030$, $RMSEA = 0.083$, $AIC = 27{,}753$, $BIC = 27{,}835$, $\chi^2(13) = 162.719$, $p < 0.001$; Wave 3: $CFI = 0.977$, $TLI = 0.963$, $SRMR = 0.037$, $RMSEA = 0.086$, $AIC = 18{,}286$, $BIC = 18{,}361$, $\chi^2(13) = 123.423$, $p < 0.001$). The two-factor models fit the data better than the one-factor models across waves (Wave 1: $\chi^2(14) = 1212.550$, $p < 0.001$; Wave 2: $\chi^2(14) = 647.978$, $p < 0.001$; Wave 3: $\chi^2(14) = 569.770$, $p < 0.001$). The two factors were moderately to highly correlated (Wave 1: $r = 0.73$; Wave 2: $r = 0.64$; Wave 3: $r = 0.55$).

We created two scales of collective action by aggregating the four *non-violent disruptive actions* items (Wave 1: $\alpha = 0.89$, Wave 2: $\alpha = 0.90$, Wave 3: $\alpha = 0.90$) and the three *violent disruptive actions* items (Wave 1: $\alpha = 0.81$, Wave 2: $\alpha = 0.80$, Wave 3: $\alpha = 0.80$).

5.2.2 | Anger Towards the Police

Anger towards the police was measured with a single item: 'Thinking of the police in general, to what extent do you feel anger?' (1 = not at all; 5 = a lot).

5.2.3 | Perceptions of Procedural Justice of the Police

Six items measured perceptions of procedural justice of the police during protests. Participants were asked, 'Thinking about the work of the police maintaining order during demonstrations, how often do you think that the police ...' (a) treat protesters with dignity and respect? (b) when they arrest a protester, do they clearly explain the reasons for arresting them? (c) when they arrest a protester, do they do it for justified reasons? (d) when dealing with protesters, do they act impartially, without differentially treating them based on social class or ethnicity? (e) do they act according to the law? (f) do they respect the protocols for detention and use of force? (1 = *never*; 5 = *always*). The scale had high reliability across all waves (αs = 0.96–0.97).

5.3 | Data Analytical Strategy

To explore the over-time associations between procedural justice, anger, and the two types of disruptive collective action measured (i.e., non-violent and violent), we employed Random-Intercept Cross-Lagged Panel Models (RI-CLPMs) using Full Information Maximum Likelihood (see general model in Figure 2). We ran the same RI-CLPM with each type of action in separate models. We modelled the associations between procedural justice, anger, and justification of violent disruptive actions in Model 1 and between procedural justice, anger, and non-violent disruptive actions in Model 2.

RI-CLPM accounts for stability and change in collective action over time, as it partitions within- and between-person variance and accurately estimates within-person longitudinal associations between variables. Stable between-person variance in each construct is captured by a separate trait factor (i.e., random intercept) (Hamaker et al. 2015). Random intercepts for each construct represent participants' average level of the construct across three measurements (e.g., participants' average level of procedural justice over the three waves) and account for the between-person stability of the analysed constructs. The three random intercepts were allowed to correlate, and the factor loadings connecting the random intercepts with the corresponding observed variables were set to 1. The correlations between random intercepts represent the between-person relations between variables averaged across time points.

Regarding the within-person part of these models, we fixed the residual variances of the observed variables to 0 to decompose the variance in the observed variables. The previous step allowed us to differentiate between a between-person component (the random intercepts) and a within-person component (latent within-person deviation scores). The autoregressive and cross-lagged effects in the RI-CLPM are modelled based on residualised scores. A positive autoregressive effect examines whether within-person deviations from the trait level of a construct persist from one measurement time to the next (Hamaker et al. 2015; Usami et al. 2019). Significant cross-lagged effects, which are the effects of interest in this paper, indicate the extent to which within-person deviations from the expected score on a construct at one time point predict within-person deviations from the individual's

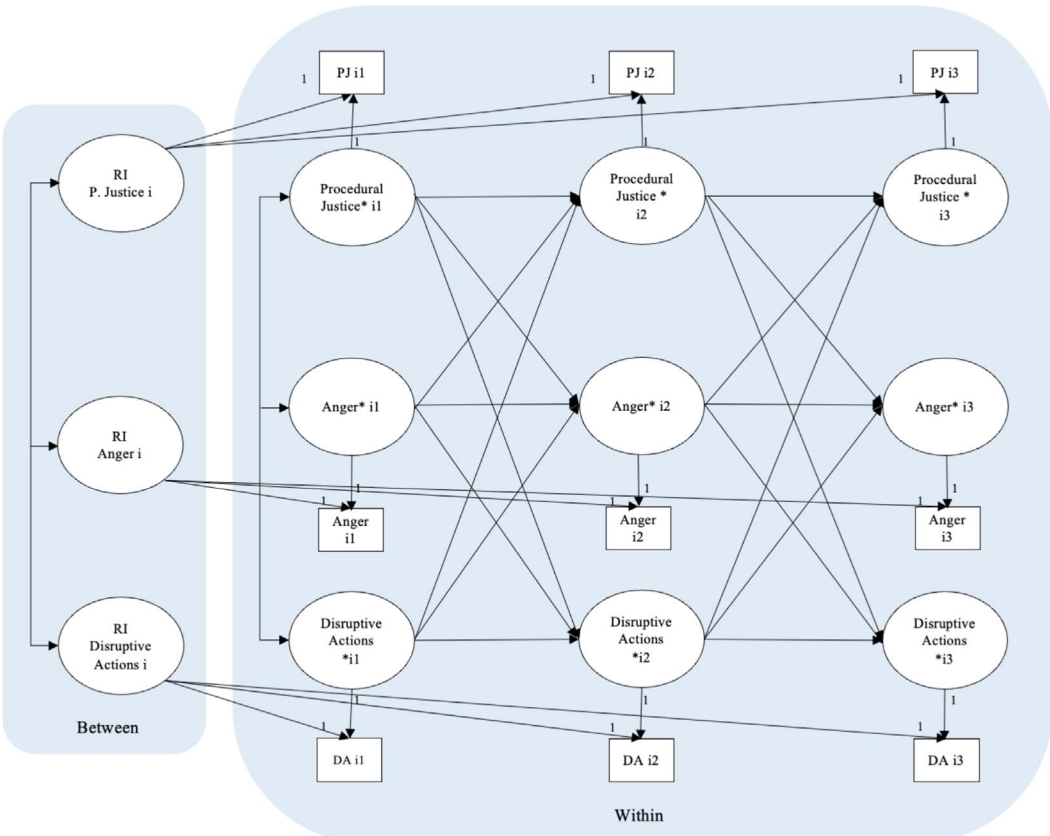

FIGURE 2 | Theoretical RI-CLPM. Procedural justice, anger and support for disruptive collective actions (in general, later violent disruptive actions and non-violent disruptive actions are tested separately). DA = disruptive actions. R.I = random intercept.

TABLE 1 | Means, standard deviations and correlations of procedural justice, anger and support for non-violent and violent actions across three waves.

Variable	*M*	SD	1	2	3	4	5	6	7	8	9	10	11
1. P. Justice w1	2.31	0.96											
2. P. Justice w2	2.49	0.96	0.87***										
3. P. Justice w3	2.64	0.99	0.85***	0.90***									
4. Anger w1	2.93	1.41	−0.72***	−0.71***	−0.68***								
5. Anger w2	2.66	1.41	−0.68***	−0.72***	−0.69***	0.78***							
6. Anger w3	2.43	1.36	−0.65***	−0.67***	−0.68***	0.73***	0.77***						
7. Non-V DA w1	2.18	0.88	−0.66***	−0.68***	−0.68***	0.62***	0.59***	0.65***					
8. Non-V DA w2	2.13	0.87	−0.66***	−0.67***	−0.70***	0.62***	0.60***	0.66***	0.82**				
9. Non-V DA w3	2.07	0.85	−0.65***	−0.68***	−0.69***	0.60***	0.60***	0.66***	0.80**	0.84***			
10. Viol DA w1	1.47	0.66	−0.51***	−0.50***	−0.52***	0.54***	0.52***	0.57***	0.67**	0.61***	0.62***		
11. Viol DA w2	1.41	0.58	−0.53***	−0.54***	−0.56***	0.55***	0.55***	0.59***	0.63**	0.69***	0.66***	0.75***	
12. Viol DA w3	1.34	0.54	−0.48***	−0.46***	−0.49***	0.49***	0.50***	0.56**	0.59***	0.59***	0.66**	0.73***	0.77***

Note: M and SD are used to represent mean and standard deviation, respectively.
Abbreviations: P. Justice = procedural justice; Non-V DA = non-violent disruptive actions; Viol DA = violent disruptive actions.
*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

mean for another variable at a later time point (see Osborne and Little 2024). For example, a negative cross-lagged effect from perceived procedural justice to support for violent action would show that an individual's deviation from their typical mean-level perceived procedural justice at one time point is associated with lower-than-usual support for violent collective action later.

All statistical analyses were performed in RStudio using the maximum likelihood (ML) estimator and the *lavaan* package (Rosseel 2012). The adequacy of the model fit was assessed based on established criteria (Hu and Bentler 1999), considering a cut-off value close to 0.95 for TLI, CFI; a cutoff value close to 0.08 for SRMR; and a cutoff value of 0.06 for RMSEA. To examine if the data exhibited consistent individual changes over time, we compared the non-stationary model, where all parameters are free, with the stationary (constrained) model. The stationary model imposes restrictions by making the within-person autoregressive and cross-lagged effects between Wave 1 and Wave 2 equivalent to those observed between Wave 2 and Wave 3. If the non-stationary model demonstrates a better fit, it implies that within-person processes vary over time (Cole and Maxwell 2003; Hamaker et al. 2015). We present the results of the model that has the superior fit below.

6 | Results

Table 1 shows the means, standard deviations, and correlations between all variables across the three waves. Perceptions of procedural justice were slightly higher than the midpoint of the scale, whereas the levels of reported anger were at the midpoint of the scale. Support for violent actions was low, and support for non-violent disruptive actions was just below the middle of the scale. Procedural justice was negatively correlated with anger and both types of action, whereas anger and collective actions were positively correlated.

First, we tested for stationarity of the two models to examine whether a constrained or unconstrained model fits the data better. The non-stationary models (Model 1: $\chi^2(3) = 41.163$; CFI = 0.997; RMSEA = 0.047; SRMR = 0.047; Model 2: RI-CLPM, $\chi^2(3) = 23.092$; CFI = 0.998; RMSEA = 0.048; SRMR = 0.01) fit the data better than their stationary counterparts (Model 1: χ^2 (12) = 87.263; CFI = 0.991; RMSEA = 0.047; SRMR = 0.036, $\Delta\chi^2$ (9) = 46.100, $p < 0.001$; Model 2: χ^2 (12) = 61.297; CFI = 0.995; RMSEA = 0.038; SRMR = 0.022, $\Delta\chi^2$ (9) = 38.205, $p < 0.001$). Therefore, the non-stationary models (i.e., unconstrained models) will be used to interpret the results.

6.1 | Between-Person Level Effects Across Models

Across the two models, all random intercepts, which reflect the person's averaged mean score across assessments (Osborne and Little 2024), had significant variance. This suggests that there are stable differences between individuals (i.e., trait factor) in their expected scores in perceived procedural justice, anger, and justification of violent disruptive actions (Model 1) and non-violent disruptive actions (Model 2) (see Table 2 for variances and covariances).

All covariances between the random intercepts of the observed variables (see Table 2) were significant and in the theoretically expected direction for both models: perceived procedural justice and anger towards the police were negatively related, as well as perceived procedural justice and support for non-violent actions. That is, as prior research has shown, higher perceptions of procedural justice of the police are associated with lower anger towards the police, lower support for violent disruptive actions (Model 1), and lower support for non-violent disruptive actions (Model 2). By contrast, higher anger towards the police was associated with more support for violent and non-violent disruptive actions.

6.2 | Within-Person Effects Over Time: Violent Disruptive Actions (Model 1)

Table 3 and Figure 3 present all Model 1 within-person autoregressive and cross-lagged effects.

TABLE 2 | Between-person, random intercept variances and covariances for procedural justice, anger towards the police and support for violent disruptive actions (Model 1) and non-violent disruptive actions (Model 2).

	Variables		Anger	Disruptive action
		Variances	Covariances	
Model 1	1. Procedural justice	$\sigma^2 = 0.754$, SE = 0.023, $p < 0.001$	$\sigma = -0.851$, SE = 0.003, $p < 0.001$	$\sigma = -0.286$, SE = 0.014, $p < 0.001$
	2. Anger	$\sigma^2 = 1.379$, SE = 0.055, $p < 0.001$	—	$\sigma = 0.460$, SE = 0.023, $p < 0.001$
	3. Violent disruptive action	$\sigma^2 = 0.279$, SE = 0.018, $p < 0.001$	—	—
Model 2	1. Procedural justice	$\sigma^2 = 0.746$, SE = 0.026, $p < 0.001$	$\sigma = -0.833$, SE = 0.033, $p < 0.001$	$\sigma = -0.515$, SE = 0.019, $p < 0.001$
	2. Anger	$\sigma^2 = 1.36$, SE = 0.056, $p < 0.001$	—	$\sigma = 0.709$, SE = 0.029, $p < 0.001$
	3. Non-violent disruptive action	$\sigma^2 = 0.559$, SE = 0.023, $p < 0.001$	—	—

TABLE 3 | Model 1 parameter estimates for the within-person effects (autoregressive and cross-lagged effects).

Predictor	Outcome	Estimate	SE	*p*
Procedural justice T_1	Procedural justice T_2	0.174*	0.081	0.032
Procedural justice T_2	Procedural justice T_3	0.367***	0.084	< 0.001
Anger T_1	Anger T_2	0.221**	0.070	0.002
Anger T_2	Anger T_3	0.103	0.080	0.201
Violent DA T_1	Violent DA T_2	0.147	0.078	0.057
Violent DA T_2	Violent DA T_3	0.026	0.165	0.876
Procedural justice T_1	Anger T_2	−0.336**	0.129	0.009
Procedural justice T_1	Violent DA T_2	−0.133*	0.053	0.012
Anger T_1	Procedural justice T_2	−0.139***	0.035	< 0.001
Anger T_1	Violent DA T_2	0.005	0.022	0.814
Violent DA T_1	Procedural justice T_2	−0.069	0.077	0.370
Violent DA T_1	Anger T_2	−0.096	0.122	0.432
Procedural justice T_2	Anger T_3	−0.103	0.142	0.470
Procedural justice T_2	Violent DA T_3	0.150	0.081	0.064
Anger T_2	Procedural justice T_3	−0.025	0.036	0.490
Anger T_2	Violent DA T_3	−0.066*	0.033	0.048
Violent DA T_2	Procedural justice T_3	−0.186	0.107	0.081
Violent DA T_2	Anger T_3	0.199	0.216	0.358

Note: Violent disruptive actions (non-stationary model).
*$p < 0.05$. **$p < 0.01$. ***$p < 0.001$.

6.2.1 | Autoregressive Effects

The autoregressive paths of procedural justice were significant ($\beta_{T1-T2} = 0.174$, SE = 0.081, $p = 0.032$; $\beta_{T2-T3} = 0.367$, SE = 0.084, $p = 0.000$). For anger, only the autoregressive path from Time 1 to Time 2 was significant ($\beta_{T1-T2} = 0.221$, SE = 0.07, $p = 0.002$; Anger $\beta_{T2-T3} = 0.103$, SE = 0.080, $p = 0.201$). The autoregressive paths for violent actions were not significant ($\beta_{T1-T2} = 0.147$, SE = 0.078, $p = 0.057$; $\beta_{T2-T3} = 0.026$, SE = 0.165, $p = 0.876$).

6.2.2 | Cross-Lagged Effects

In line with H1, cross-lagged effects between procedural justice and anger show a significant negative association between procedural justice at T1 and anger at T2 ($\beta_{T1-T2} = -0.336$, SE = 0.129, $p = 0.009$), but no significant association between procedural justice at T2 and anger at T3 ($\beta_{T1-T2} = -0.103$, SE = 0.142, $p = 0.470$). In addition, higher anger at T1 predicted lower procedural justice at T2 ($\beta_{T1-T2} = -0.139$, SE = 0.035, $p = 0.000$), but this association was non-significant between T2 and T3.

Importantly and supporting H2, participants who perceived more procedural justice than thei average in T1 expressed less support for violent disruptive actions in T2 ($\beta_{T1-T2} = -0.133$, SE = 0.053, $p = 0.012$). Still, this effect was not significant between T2 and T3 ($\beta_{T1-T2} = 0.150$, SE = 0.081, $p = 0.064$). Contrary to our expectations on H3, anger at T1 did not predict support for violent actions at T2 ($\beta_{T1-T2} = 0.005$, SE = 0.022, $p = 0.814$). Still, surprisingly, participants who reported greater anger towards the police in T2 were *less* supportive of violent disruptive actions in T3 ($\beta_{T1-T2} = -0.066$, SE = 0.033, $p = 0.048$). There were no significant cross-lagged effects from violent actions to either procedural justice or anger over time.

To sum up, beyond the bidirectional cross-lagged effects between perceived procedural justice and anger from Time 1 to Time 2 (H1), in this model, higher-than-average levels of police procedural justice were associated with lower support for violent disruptive actions in the same period (H2). Surprisingly and different from what we expected (H3), from Time 2 to Time 3, the only cross-lagged effect was from increased anger towards the police to lower support for violent disruptive actions.

6.3 | Within-Person Effects Over Time: Non-Violent Disruptive Actions (Model 2)

Table 4 and Figure 4 present Model 2 within-person autoregressive and cross-lagged effects.

6.3.1 | Autoregressive Effects

For procedural justice, the path from T1 to T2 was not significant ($\beta_{T1-T2} = 0.013$, SE = 0.091, $p = 0.152$), but the autoregressive path from T2 to T3 was significant ($\beta_{T2-T3} = 0.335$, SE = 0.087,

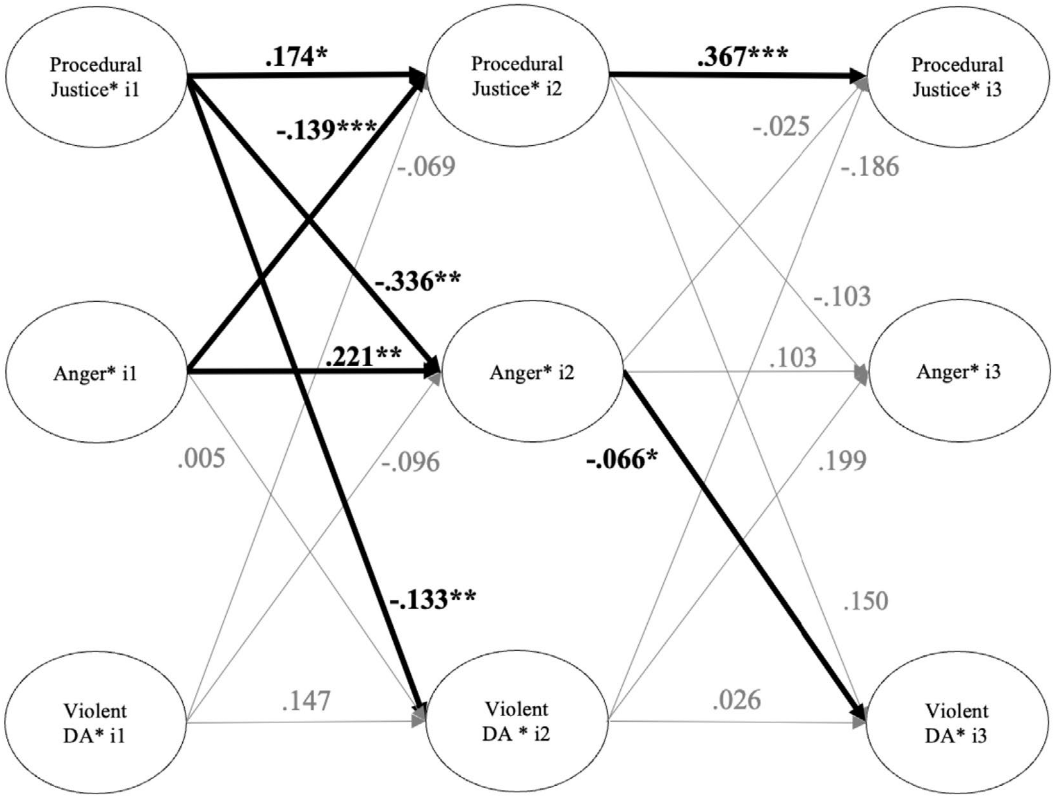

FIGURE 3 | Within-person longitudinal associations for procedural justice, anger toward the police and violent disruptive actions (non-stationary random intercept cross-lagged panel model). Grey dashes represent non-significant effects. Bold black lines represent statistically significant paths. Violent DA = violent disruptive actions.

$p = 0.000$). For anger, the autoregressive path from T1 to T2 was significant ($\beta_{T1\text{–}T2} = 0.205$, SE = 0.070, $p = 0.003$), but not from T2 to T3 ($\beta_{T2\text{–}T3} = 0.098$, SE = 0.079, $p = 0.216$). Both autoregressive paths in non-violent disruptive actions were significant ($\beta_{T1\text{–}T2} = 0.267$, SE = 0.078, $p = 0.001$; $\beta_{T2\text{–}T3} = 0.226$, SE = 0.081, $p = 0.005$).

6.3.2 | Cross-Lagged Effects

Similar to the previous model, an examination of cross-lagged effects showed that higher levels of procedural justice at Time 1 predicted lower levels of anger (H1) at Time 2 ($\beta_{T1\text{–}T2} = -0.309$, SE = 0.134, $p = 0.022$). Still, this association did not hold from Time 2 to Time 3 (see Table 3 and Figure 3). Importantly, within-person deviations in procedural justice at Time 1 and Time 2 did not predict support for non-violent action at subsequent time points (i.e., Time 2 and Time 3, respectively).

Participants' greater levels of anger towards the police at Time 1 were associated with a subsequent decrease in their perceptions of procedural justice of the police at Time 2 ($\beta_{T1\text{–}T2} = -0.127$, SE = 0.037, $p = 0.001$). However, anger at Time 2 was not associated with procedural justice in Time 3. In this model, H3 was not supported; anger at Time 1 and Time 2 was not associated with support for non-violent actions in subsequent time points ($\beta_{T1\text{–}T2} = 0.008$, SE = 0.033, $p = 0.814$; $\beta_{T2\text{–}T3} = -0.066$, SE = 0.038, $p = 0.079$). Finally, support for non-violent actions at Time 1 and Time 2 did not predict anger ($\beta_{T1\text{–}T2} = -0.182$, SE = 0.110, $p = 0.099$) or procedural justice ($\beta_{T1\text{–}T2} = -0.052$, SE = 0.065, $p = 0.426$).

To sum up, the only cross-lagged effects in Model 2 included a bi-directional influence between procedural justice and anger from Time 1 to Time 2 (H1), which was also observed in Model 1. No other cross-lagged effects were significant. That is, neither procedural justice of the police (H2) nor anger towards police (H3) predicted justification of non-violent disruptive actions over time.

7 | Discussion

Responding to calls for the employment of longitudinal methods and analysis in collective action (Górska and Tausch 2022; Hässler et al. 2022), this study examined the relationships between procedural justice, emotions, and violent and non-violent disruptive actions longitudinally during a dynamic socio-political environment. We employed Random Intercept Cross-Lagged Panel Models (RI-CLPM) to investigate the procedural justice beliefs, anger towards the police, and support for collective action of Chilean citizens at three time points. Notably, the results revealed different associations between constructs over the two time periods marked by significant socio-political promises of change. Therefore, we will interpret the results separately for the periods from Time 1 to Time 2 and from Time 2 to Time 3, situating them within the sociopolitical context of the country.

Higher perceived procedural justice in Time 1 (following the decision to change the constitution) was associated with lower

TABLE 4 | Model 2 parameter estimates for the within-person effects (autoregressive and cross-lagged effects).

Predictor	Outcome	Estimate	SE	*p*
Procedural justice T_1	Procedural justice T_2	0.013	0.091	0.152
Procedural Justice T_2	Procedural justice T_3	0.335***	0.087	< 0.001
Anger T_1	Anger T_2	0.205**	0.070	0.003
Anger T_2	Anger T_3	0.098	0.079	0.216
Non-violent DA T_1	Non-violent DA T_2	0.267***	0.078	0.001
Non-violent DA T_2	Non-violent DA T_3	0.226**	0.081	0.005
Procedural Justice T_1	Anger T_2	−0.309	0.134	0.022*
Procedural Justice T_1	Non-violent DA T_2	−0.030	0.074	0.683
Anger T_1	Procedural justice T_2	−0.127***	0.037	0.001
Anger T_1	Non-violent DA T_2	0.008	0.033	0.814
Non-violent DA T_1	Procedural Justice T_2	−0.052	0.065	0.426
Non-violent DA T_1	Anger T_2	−0.182	0.110	0.099
Procedural Justice T_2	Anger T_3	−0.020	0.138	0.885
Procedural Justice T_2	Non-violent DA T_3	−0.018	0.087	0.838
Anger T_2	Procedural justice T_3	−0.020	0.038	0.609
Anger T_2	Non-violent DA T_3	−0.066	0.038	0.079
Non-violent DA T_2	Procedural justice T_3	−0.082	0.066	0.217
Non-violent DA T_2	Anger T_3	0.165	0.127	0.192

Note: Non-violent disruptive actions (non-stationary model).
Abbreviation: Non-violent DA = non-violent disruptive actions.
$^*p < 0.05$. $^{**}p < 0.01$. $^{***}p \leq 0.001$.

justification of violent disruptive actions in Time 2 (after the commission to write the constitution draft was elected). In other words, people who viewed the police as acting more fairly were later less supportive of violent disruptive actions, such as damaging public property or throwing stones at the police. This result aligns with previous cross-sectional research, which shows that fair treatment by authorities can enhance legitimacy and compliance (Sunshine and Tyler 2003) and reduce support for violent protest tactics (Jackson et al. 2013; Snipes et al. 2019; Tyler et al. 2018).

The results also revealed dynamic bi-directional associations between procedural justice and anger from Time 1 to Time 2 in both models. Participants with higher perceptions of police fairness in Time 1 expressed less anger towards the police in Time 2. This is consistent with prior research findings (e.g., Maguire et al. 2023). Notably, the opposite direction was also significant, such that anger was also an antecedent of perceived injustice. That is, people who were angrier at the police in Time 1 reported lower perceptions of procedural justice in Time 2.

Following the police violence in the early wave of the 2019 protests, justice and reparation measures to address the police's excessive use of force during the demonstrations were announced, but they were not implemented at that time. These unaddressed grievances might in themselves be viewed as a continuation of injustice. The continued failure to implement policies and measures to address past police violence in Time 1 may have shaped and influenced both anger and perceived fairness of police actions (i.e., procedural justice) in Time 2. In other words, those who experienced increased anger at the police in Time 1 later experienced increased anger and a higher perception of the police acting unfairly because there were no concrete changes in the police and their accountability for the human rights violations during the outbreak.

Neither perceptions of fairness nor anger towards the police were associated with support for non-violent disruptive actions. Still, only those participants who already supported non-violent disruptive actions at higher levels expressed increased and stable support for these actions in the following period (autoregressive effects). This may be explained by Chile's historical tradition of collective action, where non-violent disruptive actions are seen as the norm and have been common since at least the first student revolution in 2006 (Mayorga 2024). In other words, people's perceptions or emotions towards the police did not impact support for non-violent actions, indicating that individuals who already support these actions (e.g., protesting without authorization or barricading the street) will likely continue to do so, even in a changing context and amid institutional shifts.

A different pattern was observed for support for violent disruptive actions. From Time 1 to Time 2, support for these actions was not associated with prior levels of support for violence, but rather with perceptions of police procedural justice. From Time 2 to Time 3, support for violent action was

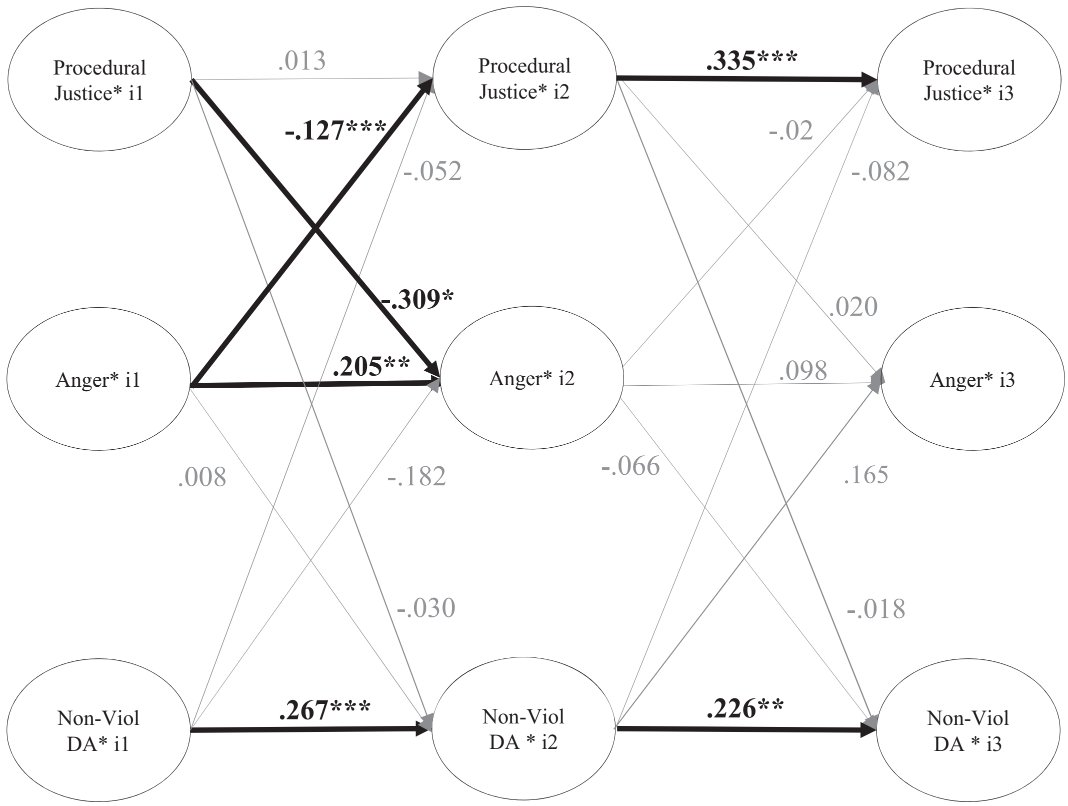

FIGURE 4 | Within-person longitudinal associations for procedural justice, anger toward the police and non-violent disruptive actions (non-stationary random intercept cross-lagged panel model). Grey dashes represent non-significant effects. Bold black lines represent statistically significant paths. Non-Viol DA = non-violent disruptive actions.

instead associated with anger, in line with previous findings (Gerber et al. 2018). These results are consistent with expectations and prior literature that link perceptions of injustice and emotional responses to support for violent protest tactics (Gerber et al. 2018).

The broader sociopolitical context may provide insight into these patterns. After the October 2020 referendum, when a majority of Chileans voted to begin drafting a new constitution, Chilean institutions appeared to adopt a more responsive stance towards ongoing protests. The December 2021 presidential election ushered in a transition from a right-wing to a progressive government under Gabriel Boric, whose proposed reforms were largely aligned with the demands raised by protesters. This political shift and the accompanying public optimism may be relevant to understanding the absence of crosslagged associations from procedural justice to anger or to support for disruptive actions during this period. As Osborne and Little (2024) point out, identifying the appropriate time lags in longitudinal research is crucial for capturing psychosocial change, which becomes especially complex in contexts marked by political transformation, where the status quo is shifting but not yet stabilised.

Surprisingly, from Time 2 to Time 3, higher anger towards the police was associated with lower support for violent disruptive actions, potentially reflecting an effect of the evolving political context. It is possible that higher anger towards the police in Time 2 did not drive support for violent disruptive actions at Time 3 because the new government headed by a leftist president, Boric, was already elected. Chile was also drafting what could have been the 'most progressive constitution in the world' (Bartlett 2022). These changes might have led those who were angrier at the police in Time 2 to be less supportive of violent action at Time 3, once a government aligned with their ideas was in power. Because collective action items in our study do not specify the goal of collective action, participants in Time 3 may have interpreted collective action as an action against the newly elected left-wing government (contrary to how it was understood earlier). As before, these findings highlight the importance of social context in understanding the predictors of support for collective action.

Another potential explanation for the association between higher anger and lower support for violent disruptive actions from Time 2 to Time 3 could be related to the increased perception of insecurity and crime following the more restrictive COVID-19 measures (Valencia 2023; INE 2022). Anger towards the police at Time 2 may be related to this heightened perception of insecurity and threat as a response to the increased crime rate, which reached a record level in 2022 (Grimaldi 2022; INE 2022; Valencia 2023). So, anger could be a response to the police's inability to handle these issues, rather than anger due to their behaviour during the protests occurring at that time (i.e., Wave 3). This explanation is plausible because the measure of anger assessed anger towards the police in general, and not specifically towards the police's behaviour during protests.

Political orientation may influence how individuals perceive and emotionally respond to perceived insecurity and institutional actions. For instance, individuals on the political right may be more likely to attribute insecurity to social disorder or government inaction rather than to police misconduct, making them less likely to express anger towards the police—even when they perceive high levels of neighbourhood insecurity. This may be particularly relevant in the Chilean context, where recent survey data suggest that right-wing Chileans report higher perceptions of neighbourhood insecurity than those with other political orientations (IPSOS 2023). However, our study was not able to assess these potential differences, as more than half of our sample did not report a political orientation (see Supporting Information). Future research should explore the role of political orientation in relation to perceptions of the police and support for collective action during transitional contexts.

Beyond anger, other emotions are likely important in this context. For example, hope is crucial in mobilising individuals to participate in collective action (Wlodarczyk et al. 2017). It might be a powerful emotion in the context of de-escalation and institutional reforms signalling future social transformation. Studied together, both positive and negative emotions might help explain expectations regarding social change for both supporters and opponents.

This study's longitudinal design offers valuable insights into the temporal dynamics between perceived procedural justice, anger, and collective action. Unlike cross-sectional studies, it captures changes over time, providing insights on how these variables interact in different phases of protests across time (Hamaker et al. 2015) and how the unfolding context impacts them. We acknowledge the limitations of our study, including the use of a non-probabilistic sample and potential biases resulting from missing data. Future research with representative samples is needed to generalise the findings (Muthén et al. 2017). Our single-item measure of anger towards the police might not capture different types of anger towards the police. Given the context, it is hard to distinguish if people were angrier at the police because of their excessive use of force or for another reason. There may also be other variables that play a significant role in the de-escalation of protests, such as the authorities' legitimacy, which we did not include in this analysis.

8 | Conclusion

In summary, this study advances our understanding of the dynamic relationships between perceived police procedural justice, anger towards the police, and justification of collective actions over time. By employing Random Intercept Cross-Lagged Panel Models (RI-CLPM), we captured within-person changes across a unique socio-political context, where institutional changes were announced and the status quo changed after a presidential election. Our findings highlight three key takeaways: First, police behaviour, particularly if they are seen as acting fairly, can play a critical role in reducing anger and delegitimizing violent protest tactics. Second, context is a key factor in justifying violent disruptive actions, whereas support for non-violent disruptive actions might be less context-dependent. Lastly, we support the claim that research on collective action must be carefully contextualised, as broader political shifts and a country's history shape individuals' attitudes and emotional responses over time.

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Ethics Statement

The ethical board of Universidad Diego Portales approved the data collection.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data and other resources are available online at https://oles2.netlify.app/study/. Analysis codes have been made publicly available via OSF and can be accessed at https://osf.io/6hgdt/?view_only=7f16239c73b948a1baeb26855705105d. The design and analysis were not pre-registered.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** casp70185-sup-0001-Tables.docx.